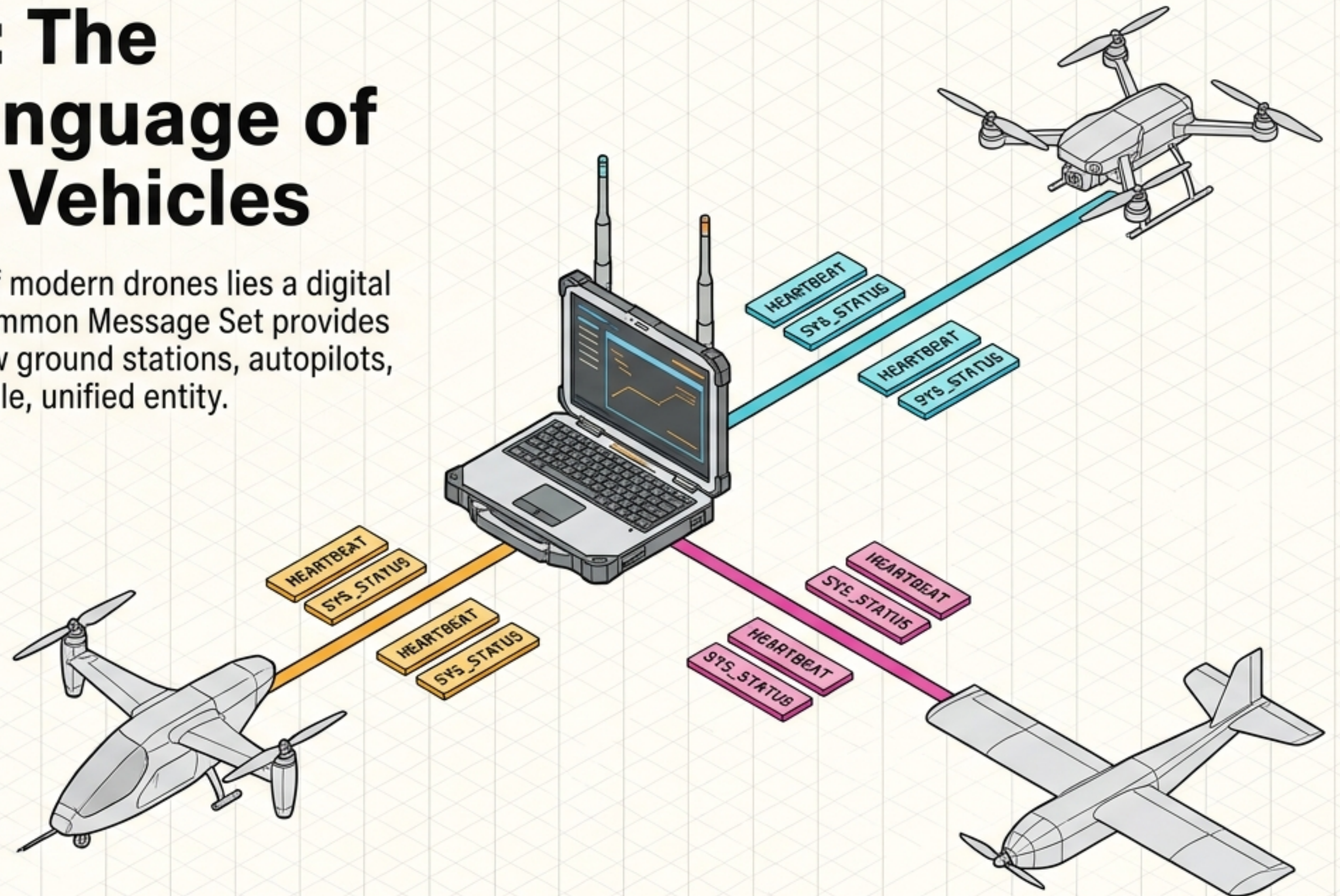


common.xml: The Universal Language of Autonomous Vehicles

Beneath the physical hardware of modern drones lies a digital nervous system. The MAVLink Common Message Set provides the standard definitions that allow ground stations, autopilots, and payloads to operate as a single, unified entity.



Protocol Snapshot

Dialect:	0
Version:	3
Defined Messages:	231

The Blueprint of Autonomy

Core Health (The Pulse)

Contains: Presence pings, system diagnostics, battery monitoring, and clock synchronization.

Spatial Senses (The Inner Ear)

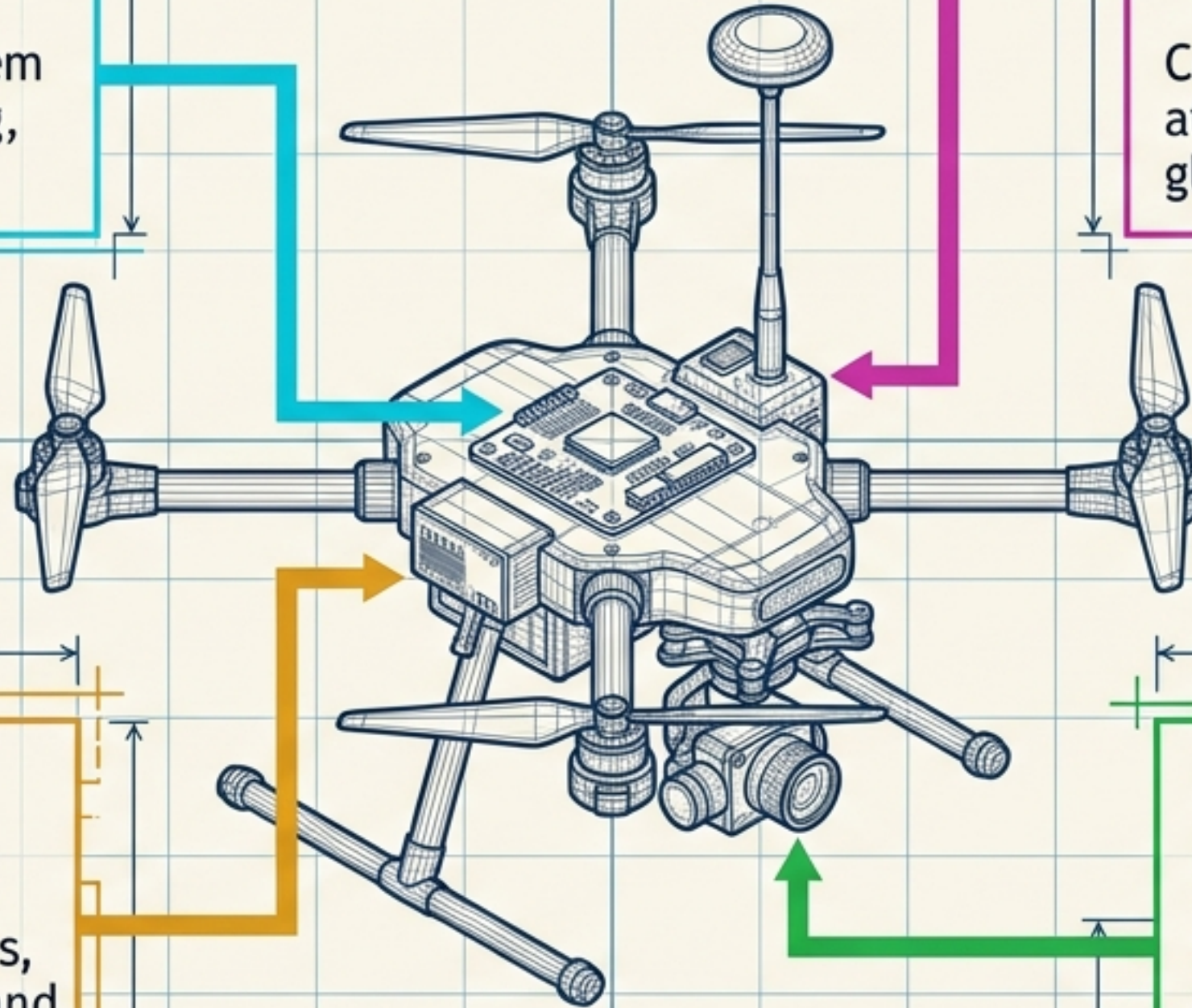
Contains: Inertial physics, atmospheric pressure, and global/local positioning matrices.

Command & Control (The Brain)

Contains: Manual stick overrides, automated mission execution, and precise coordinate commands.

Payloads & Environment (The Hands)

Contains: Optical targeting, terrain mapping, geofencing, and collision avoidance.



Every System Starts With a Heartbeat

Presence & State

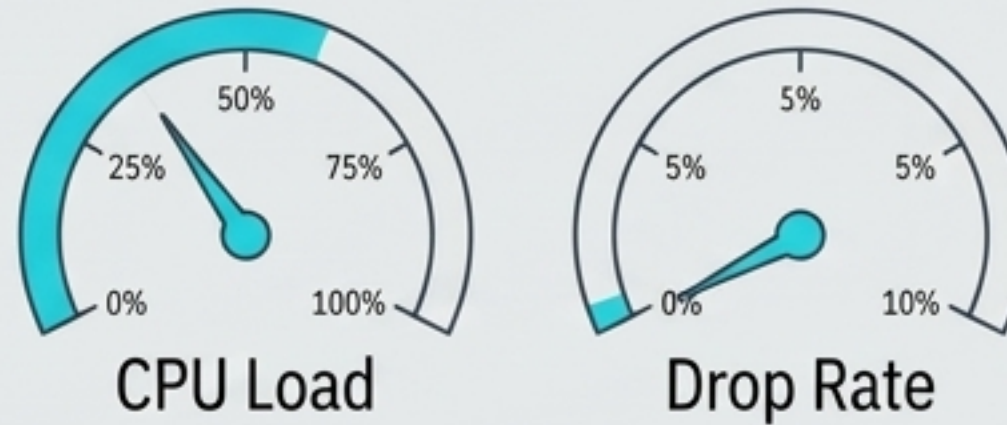


Focus: HEARTBEAT (0)

Transmits vital identity data:

- type (Vehicle/Component)
- autopilot (Flight Controller Class)
- base_mode (System mode bitmap)

Vitals & Diagnostics



Focus: SYS_STATUS (1) &
BATTERY_STATUS (147)

Key Data Points:

- load (Mainloop usage %)
- drop_rate_comm (Corrupted packets)
- current_consumed & battery_remaining

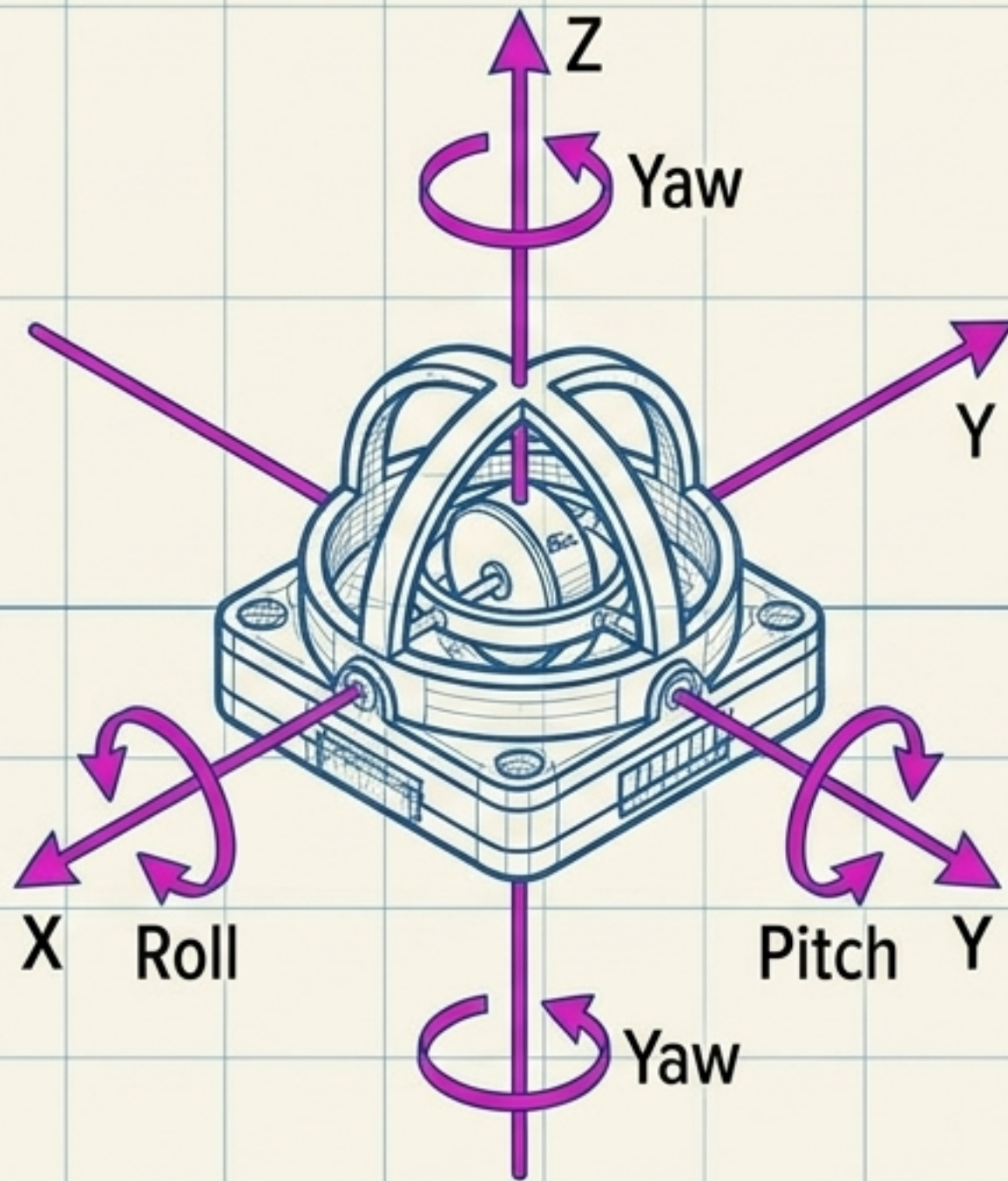
Temporal Alignment



Focus: TIMESYNC (111)

Employs timestamps tc1 (Responding component) and ts1 (Syncing component) to bridge Ground Station and vehicle clocks with nanosecond accuracy.

The Inner Ear: Inertial and Environmental Sensing



Step 1: The Raw Input

RAW_IMU (27)
RAW_PRESSURE (28)

Untampered, unscaled
ADC values (xacc,
ygyro, press_abs).
Crucial for data capture
and deep system
debugging.

Step 2: The Filtered Truth

SCALED_IMU (26)
HIGHRES_IMU (105)

Raw data converted
into standard SI units
(mG, mrad/s) and
compensated for
temperature.

Step 3: The Aeronautical State

ATTITUDE (30)

Data fused into an
aeronautical frame
(right-handed,
Z-down, X-front).
Outputs actionable
roll, pitch, and yaw in
radians.

Step 4: The Atmosphere

AIRSPEED (295) WIND_COV (231)

Calibrated Airspeed and 1-STD estimated wind vectors in XYZ.

Knowing Where You Are: The Positioning Matrix



Global Position (GNSS)

Environment: Macro, Outdoor, Global coordinates.

- GLOBAL_POSITION_INT (33): Fused global estimate (lat, lon, alt, vx, vy, vz).
- GPS_RAW_INT (24): Raw sensor data, satellites_visible, and fix_type.
- GPS_RTK (127): Millimeter precision using baseline coordinates and integer ambiguity.



Local Position (Vision & Mocap)

Environment: Micro, Indoor, GPS-denied environments.

- LOCAL_POSITION_NED (32): Fused local position in North-East-Down convention.
- VISION_POSITION_ESTIMATE (102): Camera/SLAM-based pose estimation.
- VICON_POSITION_ESTIMATE (104): Absolute tracking from external motion-capture rooms.

The Synthesis: ESTIMATOR_STATUS (230)

The arbiter. Outputs innovation test ratios to flag if a sensor measurement (like velocity or magnetometer) is rejected by the Extended Kalman Filter (EKF).

Hands on the Sticks: Translating Human Intent

The Hardware Layer (Traditional Radio)



Message:

RC_CHANNELS_RAW (35)

Mechanism: Standard PPM modulation. Measures pulse widths from 1000 microseconds (0%) to 2000 microseconds (100%).

Limit: Hardware dependent, directly tied to receiver ports.

The Software Layer (Joystick API)

Message:

MANUAL_CONTROL (69)

Mechanism: Standardized movement axes (x, y, z, r) normalized entirely in software to a range of [-1000, 1000].

Feature: Transmits up to 32 button states as individual on/off bits of a bitmask.



The Override Bridge

Message: RC_CHANNELS_OVERRIDE (70)

Function: Allows GCS software to inject fake RC values. Sending UINT16_MAX releases the channel back to the physical radio transmitter.

The Command Handshake

**Ground Station
(GCS)**

**Ground Station
(GCS)**

1. The Dispatch

The GCS chooses its messenger:

- COMMAND_LONG (76): Standard package. Sends a command ID (MAV_CMD) and up to 7 float params.
- COMMAND_INT (75): Precision package. Params 5 and 6 are scaled integers (degE7) to prevent floating-point rounding errors in spatial coordinates.

3. The Verdict

Returned immediately via COMMAND_ACK (77).

Status Outputs (MAV_RESULT):

- Accepted
- Denied
- Temporarily Rejected
- In Progress (returns 0-100% completion)

**Autopilot
(Drone)**

2. The Execution

Drone verifies target system/component ID. If matched, execution begins.

The Path Forward: Executing a Mission

1. The Definition

MISSION_ITEM_INT (73)

The modern standard for defining a waypoint. Defines the coordinate frame, scheduled action (MAV_CMD), auto-continue behavior, and precise spatial coordinates.



2. The Tracker

MISSION_CURRENT (42)

A high-frequency stream (nominally 1Hz) that constantly announces the sequence number of the active target item. Includes total items and mission_state.



3. The Arrival

MISSION_ITEM_REACHED (46)

The trigger event. Emitted the moment the drone hits the target radius. The drone evaluates the autocontinue flag to either hold position or proceed.



Target Acquisition: The Camera Protocol



Layer 1: Identification & Capabilities

CAMERA_INFORMATION (259)

Transmits hardware truths: vendor name, focal length, sensor size, and horizontal/vertical pixel resolution. Includes the URI for the camera definition file.

Layer 2: Active Control

CAMERA_SETTINGS (260), CAMERA_TRIGGER (112)

Manipulates zoom levels and focus modes. Synchronizes the exact time_usec and image sequence number between the camera and the IMU for precise photogrammetry.

Layer 3: Output & Telemetry

CAMERA_IMAGE_CAPTURED (263), VIDEO_STREAM_INFORMATION (269)

The receipt. Confirms capture success, exact GPS coordinates of the snapshot, the 3D quaternion attitude of the lens, and advertises live RTSP/TCP URIs.

Smooth Tracking: Gimbal Hierarchy

The Software Abstraction

Top Tier: The Gimbal Manager

Acts as the traffic cop. Allows a primary pilot and a secondary payload operator to safely share control of the camera without conflict.

`GIMBAL_MANAGER_SET_ATTITUDE (282)`

Accepts high-level quaternions and angular velocities. Manages primary/secondary control IDs.

The Physical Hardware

Bottom Tier: The Gimbal Device

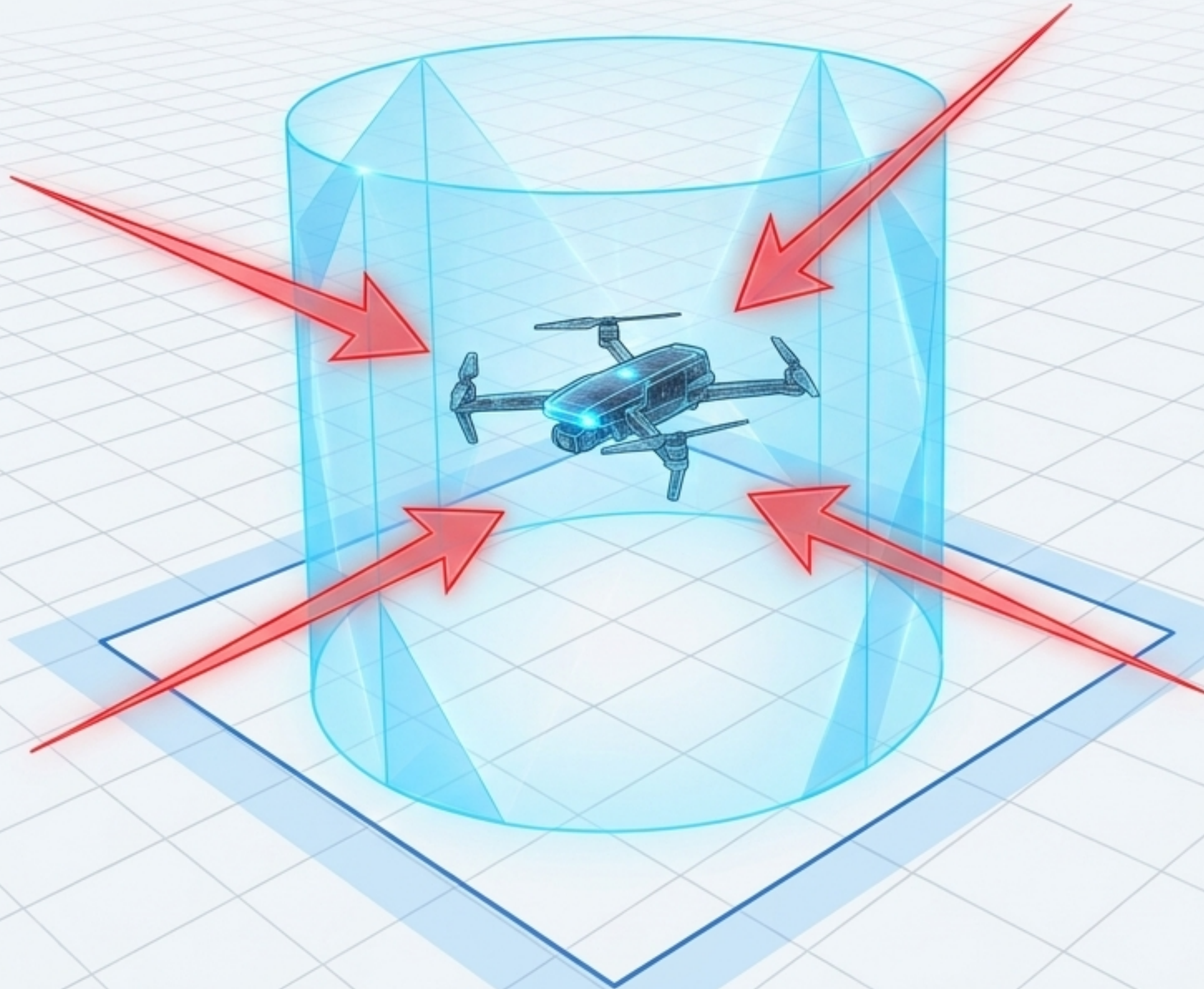
The low-level motor controller that actually moves the servos.

`GIMBAL_DEVICE_SET_ATTITUDE (284)`, `GIMBAL_DEVICE_ATTITUDE_STATUS (285)`

Deals with hardware realities: minimum/maximum mechanical pitch/yaw limits and frame-of-reference math (Earth Frame vs. Vehicle Frame).



Spatial Awareness: Fences and Threats



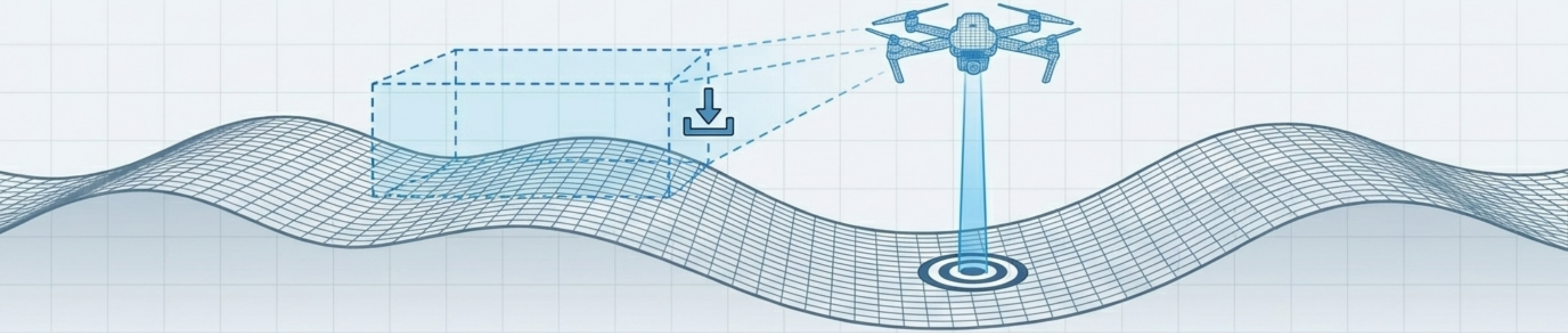
The Boundaries

- ▢ - SAFETY_SET_ALLOWED_AREA (54): Defines a strict 3D volumetric safety zone using two corners of a bounding cube.
- ✦ - FENCE_STATUS (162): The active monitor. Reports breach_count, breach_type, and the breach_mitigation action currently preventing a fly-away.

The Threats

- ▼ - ADSB_VEHICLE (246): Tracks local manned aviation. Pulls ICAO address, squawk code, heading, and altitude.
- ✦ - OBSTACLE_DISTANCE (330): A 72-element array mapping distances around the drone in specific angular increments, fed by onboard LiDAR/Radar.
- ✦ - COLLISION (247): The final warning. Evaluates threat_level and the critical time_to_minimum_delta before an impact occurs.

The Ground Below: Terrain and Ranging



1. Macro: Terrain Databases

TERRAIN_REQUEST (133), TERRAIN_DATA (134)

Downloads localized topographical maps in 4x4 grid blocks from the GCS so the drone can automatically maintain a constant height above ground level (AGL).



2. Micro: Active Clearances

DISTANCE_SENSOR (132)



Emits real-time clearance data. Tracks min_distance, max_distance, and current_distance using specific hardware orientations (e.g., ROTATION_PITCH_270 for downward-facing LiDAR).

3. The Touchdown

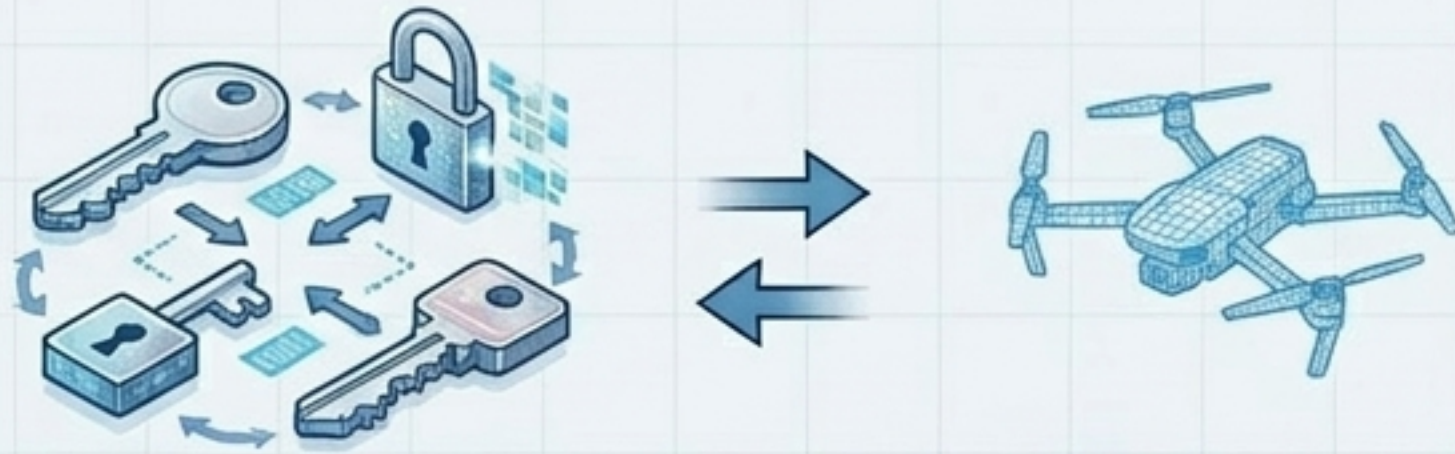
LANDING_TARGET (149)



Computer vision precision. Tracks the X/Y angular offsets and distance to a visual landing pad, calculating the exact quaternion of the target orientation to land on moving or sloped surfaces.

Deep Configuration: Parameters and Bulk Data

The Key-Value Exchange (Tuning)



Onboard parameters are stored as key -> value. GCS requests an ID, drone replies with a float.

`PARAM_REQUEST_READ (20)` → `PARAM_VALUE (22)`

The Limitation: Standard parameters are strictly limited to a 16-character ID.

`PARAM_EXT_VALUE (322)`

The Solution: Expands capabilities, allowing for 128-character string values for complex configurations.

The Heavy Lifting (Memory Extract)



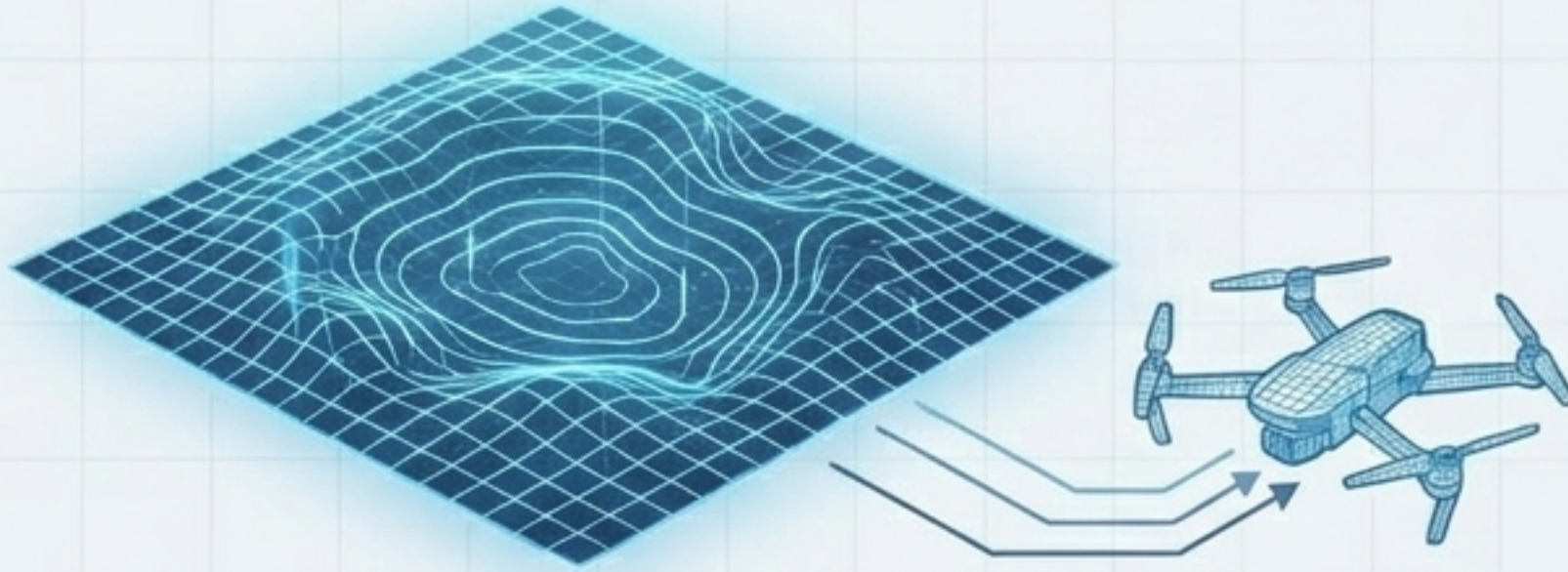
`FILE_TRANSFER_PROTOCOL (110)` →



Replaces slow sequential reads. Uses a variable length payload of up to 251 bytes per packet to efficiently move massive configuration files or compiled UAVCAN binaries over the MAVLink network.

The Extremes: Simulation and High Latency

Hardware-in-the-Loop (HIL)



Context: Fooling the physical autopilot hardware into thinking it's flying inside a software simulator.

- **HIL_STATE_QUATERNION (115):**

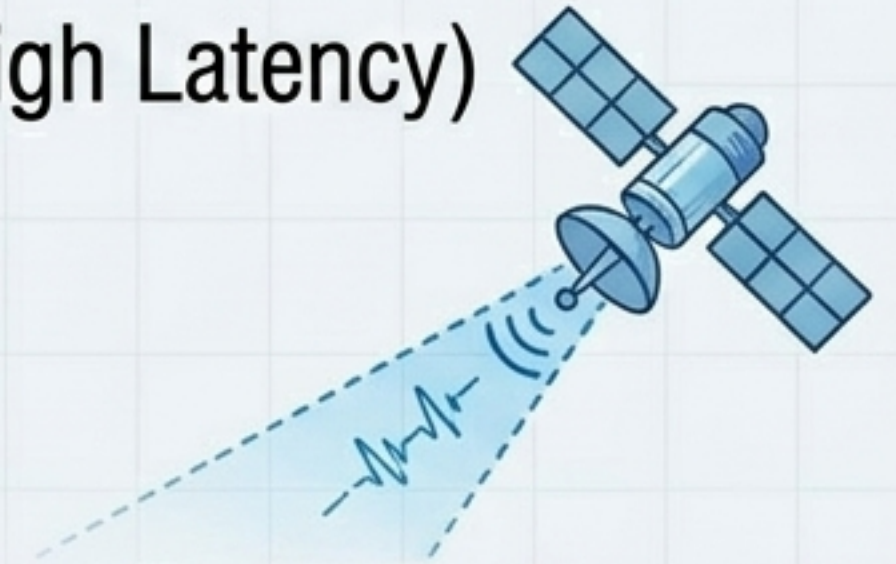
Injects simulated GPS, altitude, and ground speed.



- **HIL_SENSOR (107):** Injects fake high-resolution accelerometer and barometer physics.



Satellite Links (High Latency)

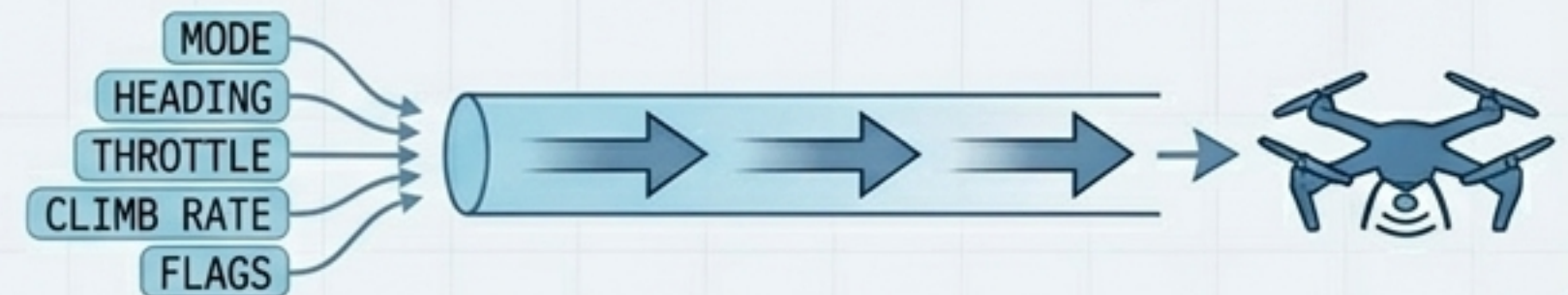


Context: Flying via Iridium satellite networks where bandwidth is incredibly expensive, tiny, and slow.

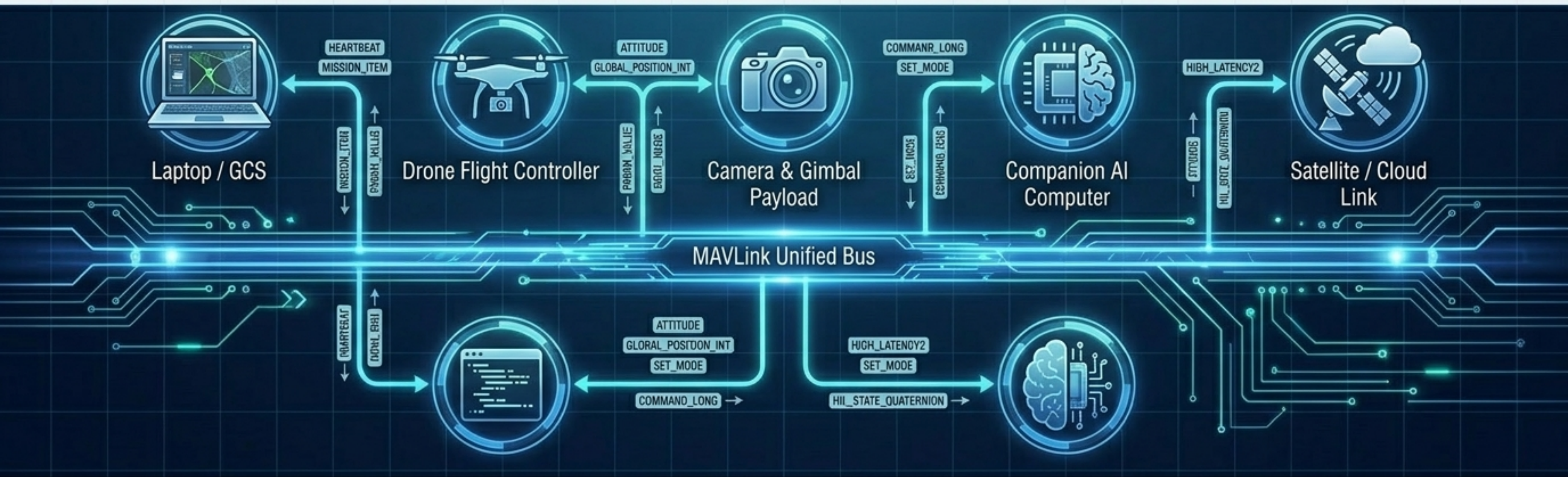
- **HIGH_LATENCY2 (235)**



Engineering: Packs only the most vital, absolute minimum data (custom mode, heading, throttle, max climb rate, and a compressed failure_flags bitmap) into a tiny payload designed to survive massive transport delays.



The Universal Language in Action



Not just a dictionary

common.xml is a highly scalable, evolving architecture that dictates system behavior, not just definitions.

Hardware Agnostic

By separating the intent (the MAVLink message) from the physical layer (motors and sensors), developers can swap components seamlessly.

The Result

A single Ground Control Station can manage an open-source mapping quadcopter, a VTOL delivery drone, and an autonomous rover—simultaneously, all speaking the exact same digital dialect.